THE UNITED REPUBLIC OF TANZANIA

REGIONAL ADMINISTRATION AND LOCAL GOVERNEMENT

FORM FOUR MOCK EXAMINATION- MANYARA.

031/2B **PHSICS 2B (actual practical)**

TIME: 2:30 HOURS **2025**

1. This paper consist of two (2) questions. Answer BOTH questions.
2. Each question carries twenty –five (25) marks
3. Non –programmable calculators may be used.
4. The following constants may be useful.
   .Acceleration due to gravity g=10mls^2
   . Pie Л=3.14

1. You are provide with a retort stand and its accessories, cotton string of 110cm, a watch, a pendulum bob, cork pads and metre rule
   (a) Set up the apparatus as shown in figure 1 below

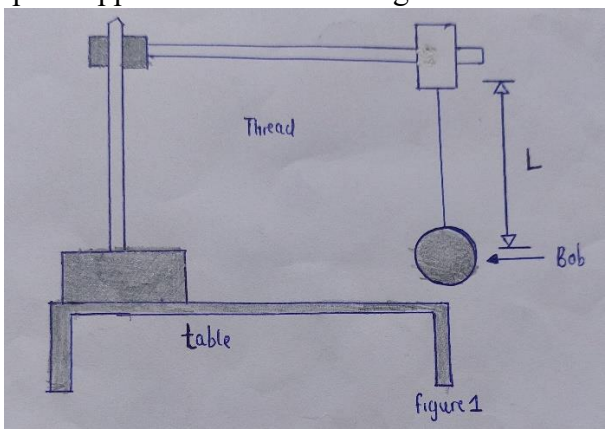


figure 1

   (b) Set the cotton string/thread at a length of a length l=10cm displace the bob and release it to make oscillation and then determine time "t" taken for complete twenty oscillations (n=20)
   (c) Repeat the procedure in (b) for the volume L=20cm, 30cm, 40cm and 50cm and then.
      i. Tabulate their results including L(cm) t (sec) and $t^2(sec^2)$
      ii. Plot on graph of $t^2(sec^2)$ a faint length L (cm)
      iii. Find the slope of the graph
      iv. Determine the value of Z, given that

$$t^2 = \frac{4Л^2n^2L}{Z}$$

      v. State the significance of Z
      vi. State the aim of this experiment.

2. The aim of this experiment is to determine the internal resistance of a cell r and e. m.f E of a dry cell D

a. You are provide with a dry cell D, resistance box (R,) an ammeter, voltmeter and key K

b. Connect the apparatuses so that you can get readings of ammeter and corresponding readings of voltmeter which was to be connect parallel to the dry cell labelled D.
c. Set the resistance box to be Iλ, read and record the ammeter and voltmeter readings.
d. Repeat the procedures for resistors R=2λ,3λ,4λ and 5λ
e. Draw a circuit diagram of connection arrangement.
f. Plot the graph of V against I
g. Write down the equation represented by the graph in terms of E, r, V and 1
h. From your graph find :
    i. The value of Electromotive force E of a dry cell D
    ii. The internal resistance r of a dry cell:
    iii. The maximum current that cell can deliver to the circuit.